

Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

DLA 19

Solute Handling
(Notes to go with online video)



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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0297. Define the composition of Na^+ in extracellular and intracellular fluid volume and its role in maintaining extracellular fluid.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0298. Calculate the normal filtered load of Na^+ . List the tubular sites of Na^+ reabsorption, and the alterations in Na^+ reabsorption in conditions of euvolemia, volume depletion, and volume expansion.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0299. Describe the regulation of Na^+ reabsorption along the nephron, including the effects of sympathetic nerves, angiotensin II, aldosterone, and atrial natriuretic peptide.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0300. Explain the formation and generation of angiotensin II, beginning with renin. Identify four factors that can promote renin release.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0301. Describe the role of the renin-angiotensin-aldosterone system in the regulation of systemic arterial blood pressure in volume-replete and volume-depleted states and in secondary forms of hypertension.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0302. Describe the role of the renin-angiotensin-aldosterone systems in the regulation of sodium balance and arterial pressure with emphasis on the actions of angiotensin II on renal hemodynamics and tubular transport.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0303. Describe K^+ distribution within the body, extrarenal K^+ homeostasis, and the role insulin, epinephrine, and aldosterone play in the movement of K^+ between intracellular and extracellular pools. Describe the K^+ shift caused by acidosis.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0304. Identify the tubular sites of K^+ reabsorption and secretion.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0305. Calculate filtered load of K^+ . Describe the factors that regulate K^+ secretion in the collecting duct

SOM.MK.II.BPM1.3.CPR.2.PHYS.0376. Explain the influence of tubular flow rate, acid-base disturbance and anion delivery on K^+ secretion

SOM.MK.II.BPM1.3.CPR.2.PHYS.0306. Identify the tubular sites of action of K^+ sparing diuretics.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0307. Identify the normal range of dietary calcium and phosphate intake. Describe the renal regulation of plasma calcium by calcitonin and phosphate by parathyroid hormone.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0308. Identify the tubular sites of Ca^{+2} and phosphate reabsorption.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0310. Describe the role of the kidney in the production of 1,25-dihydroxy vitamin D (calcitriol).

SOM.MK.II.BPM1.3.CPR.2.PHYS.0311. Describe the effects of diuretics on Ca^{+2} and phosphate excretion, especially noting the effect of thiazides to decrease Ca^{+2} excretion and loop diuretics to increase Ca^{+2} excretion.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0312. Identify the tubular sites of Mg^{+2} reabsorption and describe the regulation of Mg^{+2} by loop diuretics.



Recommended Reading

Medical Physiology: Principles for Clinical Medicine
with Companion Website Access, R. Rhoades & D. Bell
6th ed., Chapter 23.

Sodium

ECF

- Major Cation: Sodium
- Major Anions: Chloride and bicarbonate

ICF

- Major Cations: Potassium and magnesium
- Major Anions: Proteins and organic phosphate

The osmotically active solutes in the ECF are the primary determinants of ECF volume. Sodium is abundant in the ECF and makes up a major part of the osmotically active solute in the ECF. Therefore, sodium is the most important determinant of ECF volume.

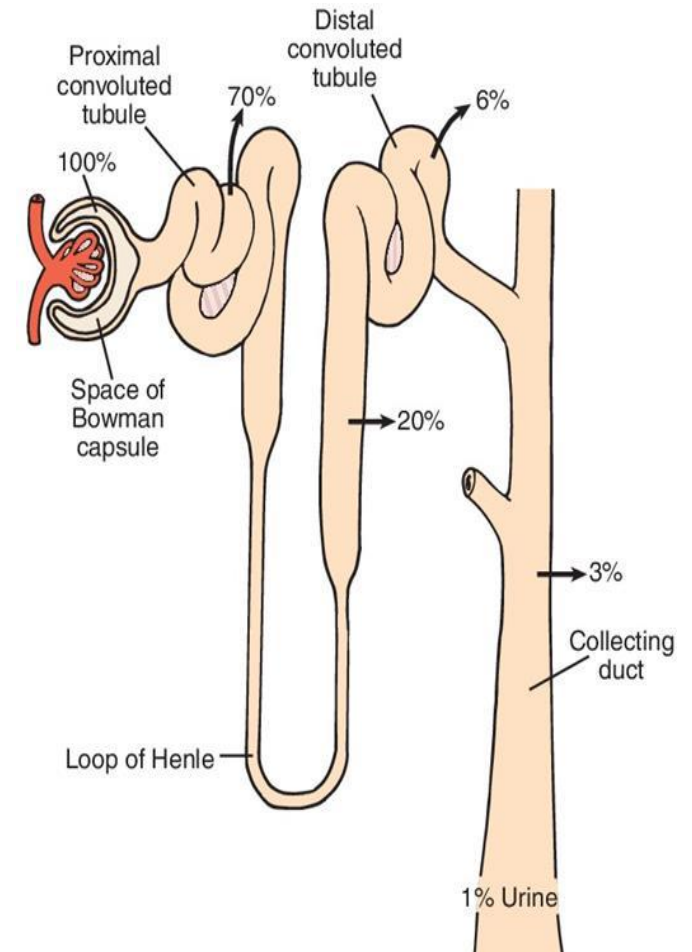
Renal Handling of Sodium

Filtered Load of Sodium: This is the amount of sodium filtered per unit time. It is dependent on the GFR and the plasma concentration of sodium.

$$FL_{Na} = GFR \times P_{[Na]}$$

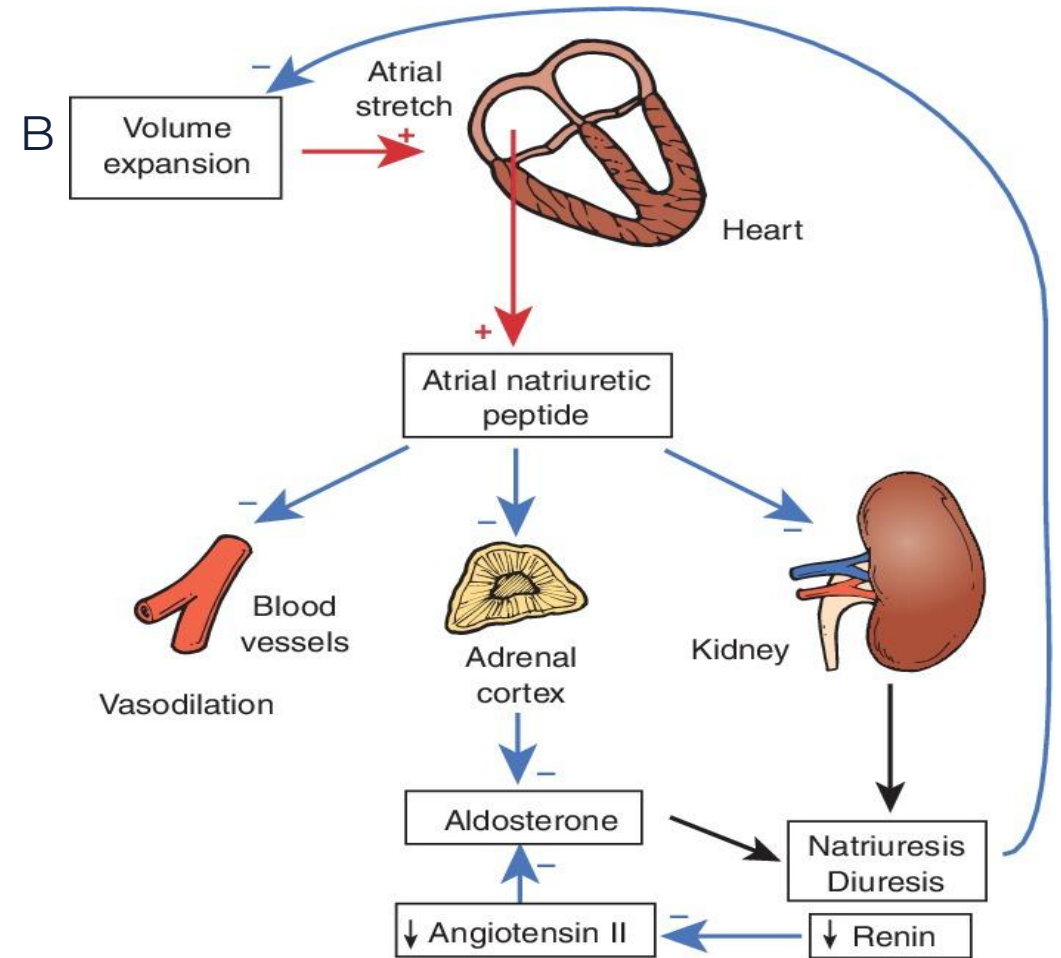
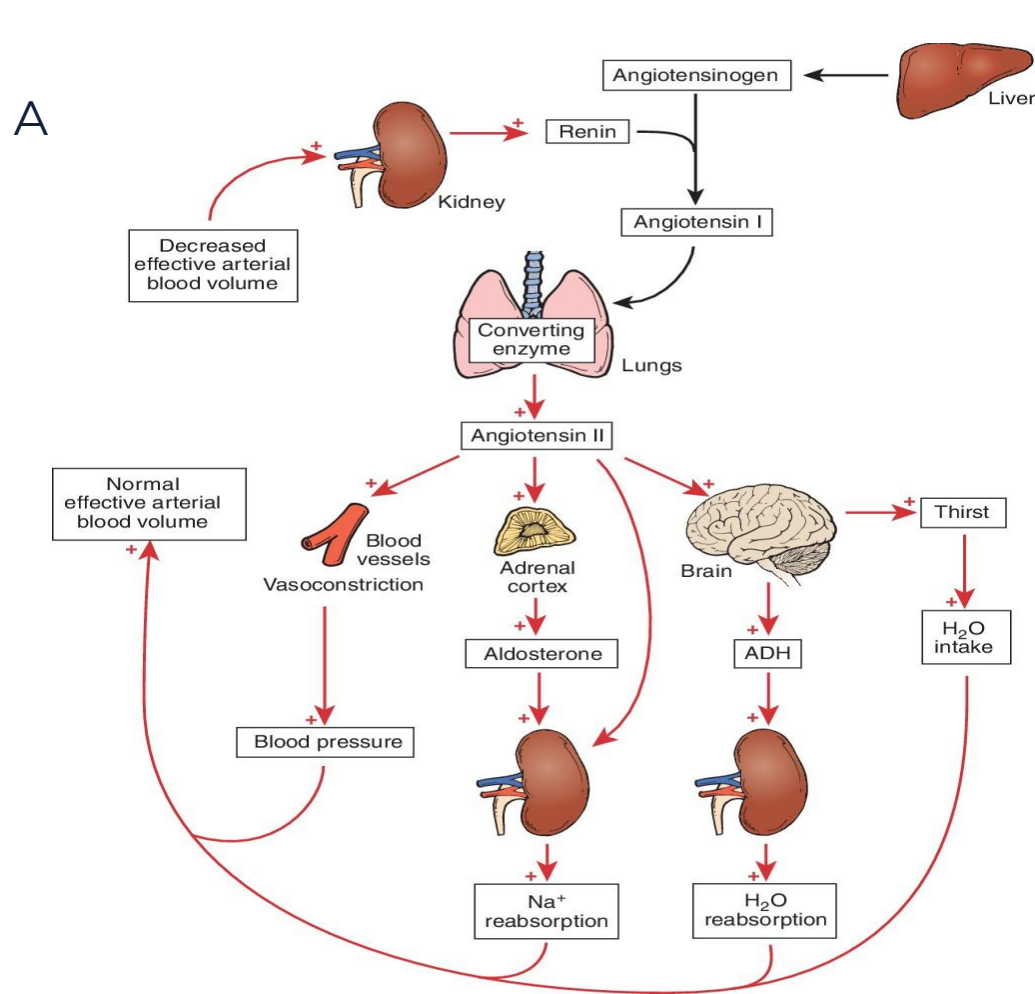
Under physiological conditions, the renal tubules reabsorb most (99%) of the filtered sodium. Reabsorption takes place in various parts of the tubules, with the descending Loop of Henle (LOH) being an exception. Approximately 1% of filtered sodium is excreted in urine.

The reabsorption rate is directly proportional to the filtered load (i.e. reabsorption increases with an increase in filtered load and decreases with a decrease in the filtered load). Reabsorption also increases in volume-depleted states and increases in the presence of volume expansion.



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Sodium Balance



A. The renin-angiotensin-aldosterone axis is integrated with the kidneys to regulate blood pressure and volume in response to a decrease in EABV

B. Stretch receptors in the heart stimulate the release of atrial natriuretic peptide (ANP). *Source: Medical Physiology: Principles for Clinical Medicine, 6e, 2023. Copyright © Wolters Kluwer.* PHYS.0299, PHYS.0300, PHYS.0301, PHYS.0302

K⁺ Distribution and Balance 1

90% of the total body K⁺ is in the ICF.

2% of K⁺ is in the ECF.

8% is found in bone, dense connective tissue, and cartilage.

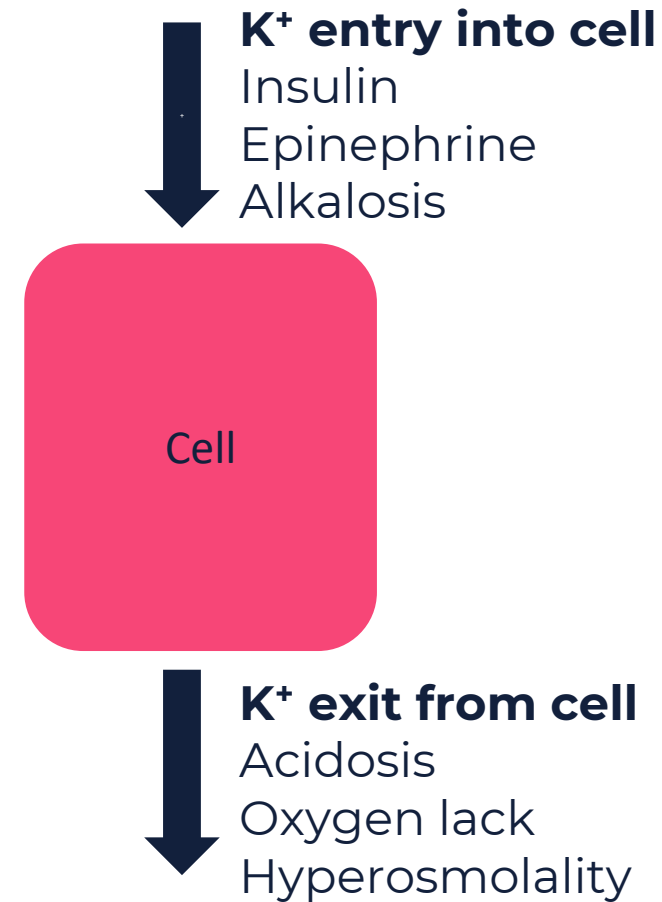
The kidneys tightly regulate plasma K⁺ (normal range 3.5 to 5.0 mEq/L).

The kidneys excrete most of the body's K⁺ (90%), with the remainder excreted in feces.

Several factors influence the distribution of K⁺ between the ECF and ICF:

Insulin, epinephrine (mediated by β_2 receptors), and increased ECF pH (alkalosis) shift K⁺ into the cell, while decreasing ECF pH (acidosis), oxygen lack, hyperosmolality, hemolysis, etc., shift K⁺ out of the cell.

In acidosis and alkalosis, the the shift of K⁺ out of the cell and into the cell respectively, is due to a passive exchange with H⁺.

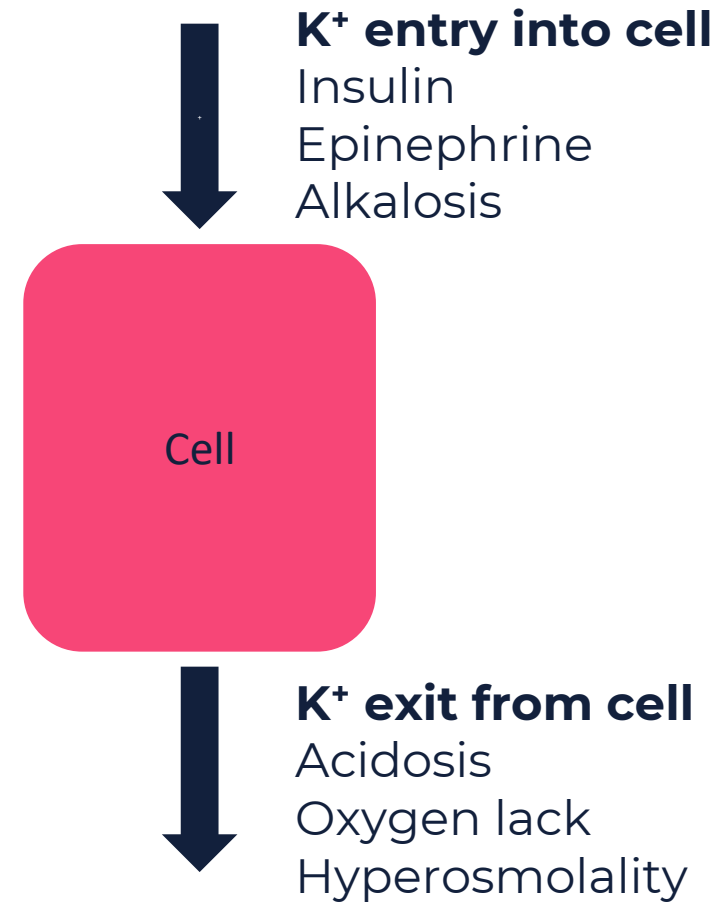


K⁺ Distribution and Balance 2

Insulin (along with glucose) can be administered in the emergency treatment of hyperkalemia.

Alkaline solutions (like NaHCO₃) can also be used for the same purpose.

Both interventions enhance K⁺ entry into the cell.



Renal Handling of K^+

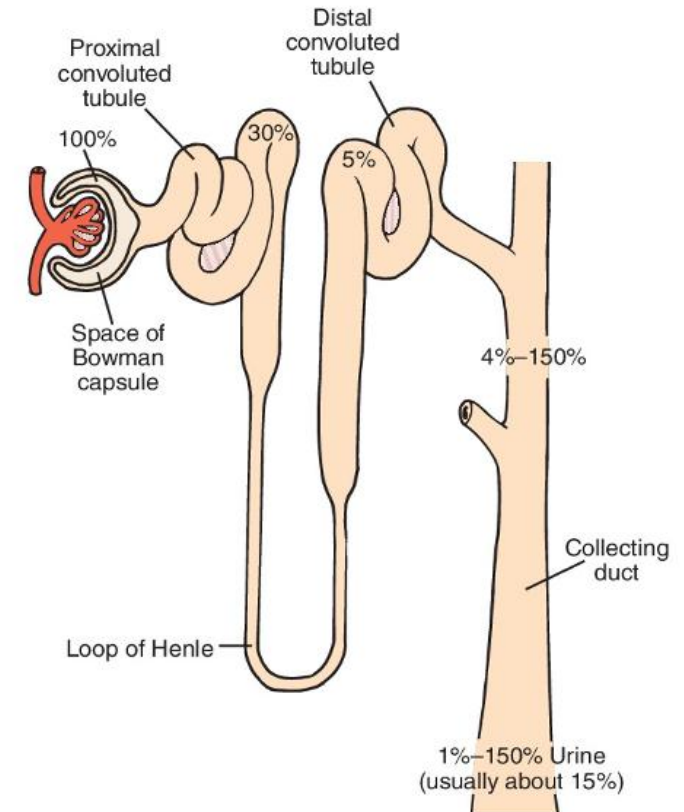
Most of the filtered K^+ is reabsorbed in the proximal convoluted tubule (70%) and the LOH (25%).

The amount of K^+ excreted is largely dependent on its secretion into the tubules by the cortical collecting ducts.

Aldosterone acts on the principal cells of the collecting ducts (increasing the activity of basolateral Na^+/K^+ -ATPase pumps, and the luminal permeability to K^+) to enhance the **secretion of K^+** into the tubules.

K^+ excreted in urine is usually about 15%, but can range from 1% to 150% depending on its levels in the body.

K^+ sparing diuretics block the action of aldosterone and epithelial Na^+ channels (ENAC), resulting in K^+ retention



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Calcium

About 1g of calcium enters the GIT daily, of which 50% comes from the diet.

40% of the calcium in the GIT is absorbed, with the remainder excreted in feces.

Most of the calcium in the body is found in bone (99%).

Very little calcium is therefore found in ECF, of which 50% is bound (to albumin and diffusible anions), and 50% is free (ionized).

Parathyroid hormone (PTH) and Vitamin D₃

Low plasma calcium stimulates the production of PTH (from the parathyroid glands) which in turn stimulates the conversion of Vit D₃ to its active form in the kidney. Active Vit D₃ ultimately increases the reabsorption of calcium in the GIT.

PTH also increases the release of calcium from bone, as well as renal reabsorption.

Renal Handling of Calcium 1

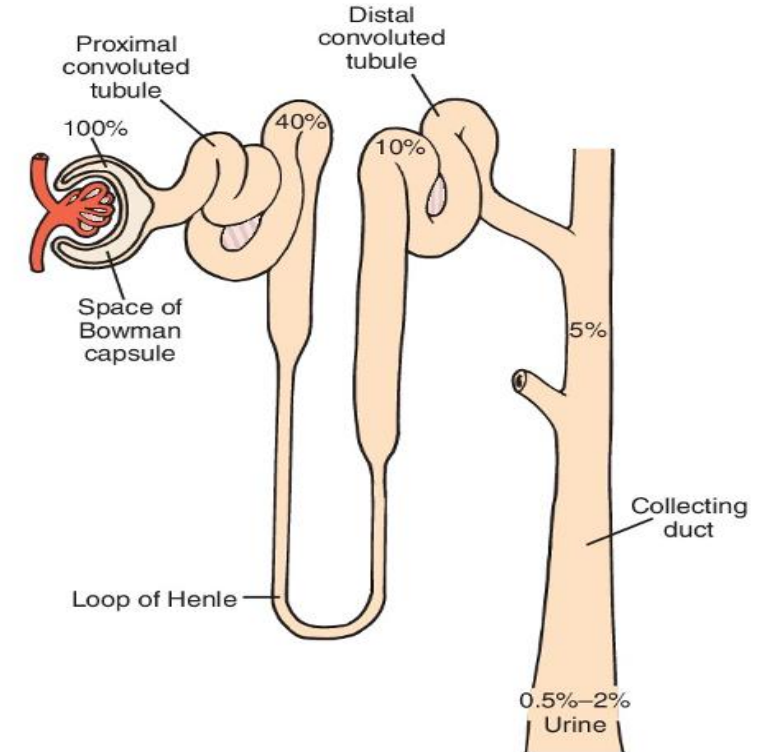
Unbound calcium is filtered at the glomeruli.

60% of the filtered calcium is reabsorbed at the PCT (paracellularly and transcellularly via apical calcium channels and basolaterally via Ca^{2+} -ATPase or $\text{Na}^+/\text{Ca}^{2+}$ exchanger).

30% is reabsorbed in the LOH.

The remainder is reabsorbed in the DCT, connecting tubules and collecting ducts.

PTH facilitates the reabsorption of calcium in the LOH and the distal tubules.



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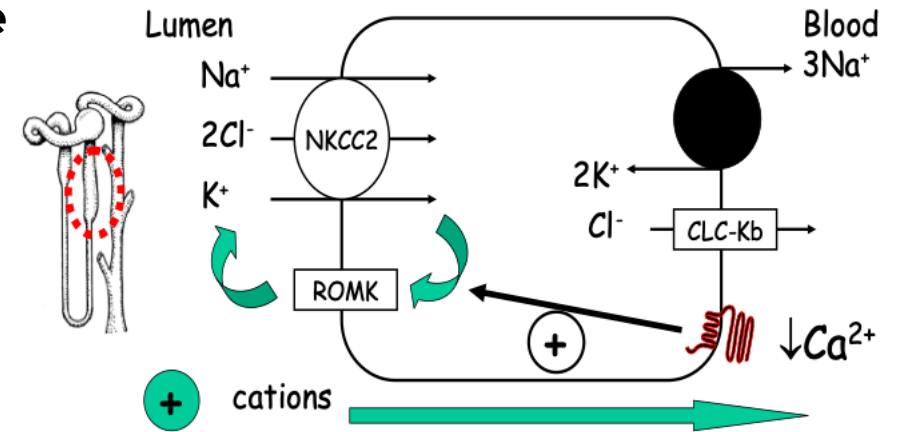
Renal Handling of Calcium 2

Loop diuretics cause increased excretion of calcium in urine

Loop diuretics, like furosemide, inhibit the action of Na-K-Cl cotransporter (NKCC) on the apical membrane in the thick ascending LOH.

NKCC activity results in the generation of a positive transepithelial potential, that drives calcium reabsorption paracellularly.

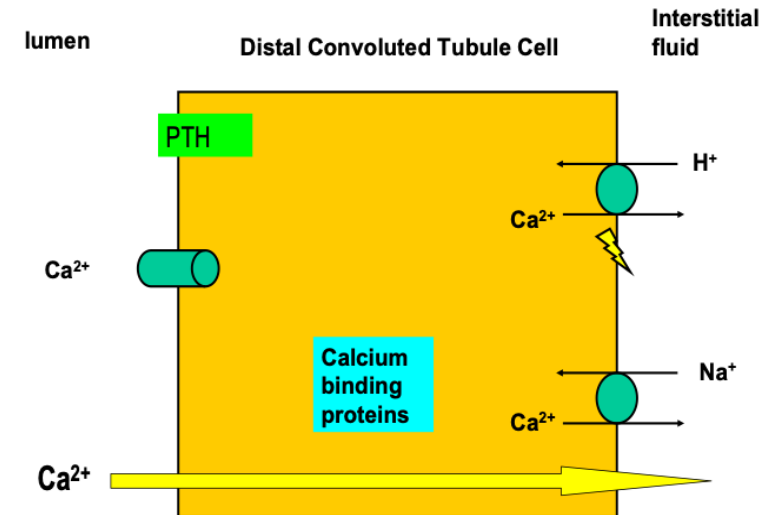
Inhibition of this activity by loop diuretics reduces calcium reabsorption and increases excretion in urine.



Thiazide diuretics cause increased reabsorption of calcium

Thiazide diuretics inhibit the action of sodium-chloride cotransporter (NCC) on the apical membrane of the DCT.

The inhibition of NCC activity increases Na/Ca exchanger activity on the basolateral membrane, which moves more sodium into the cell in exchange for calcium moving into the blood. This further increases the reabsorption of calcium from the lumen and decreases its excretion.



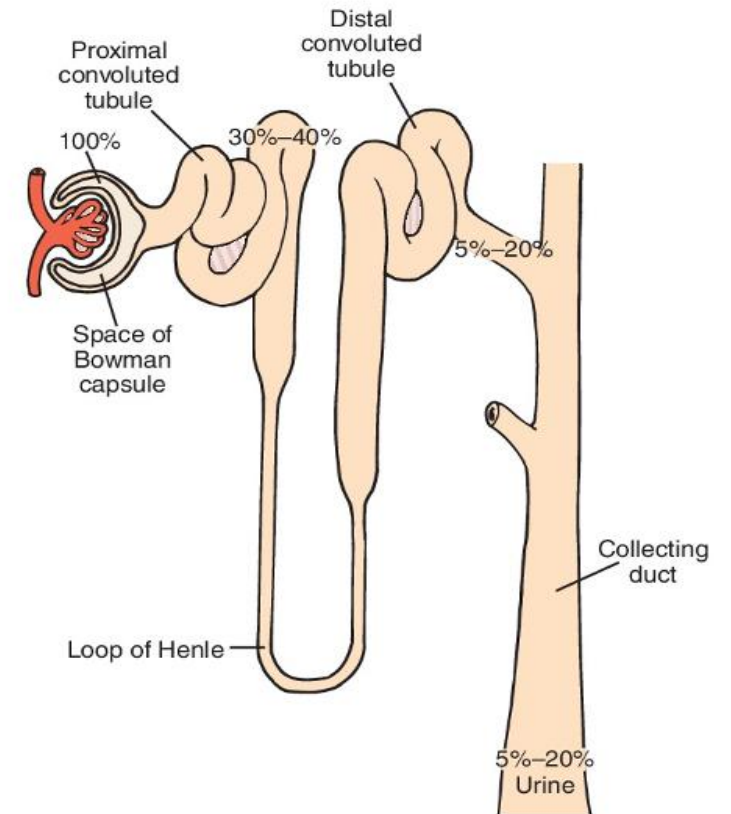
Phosphate Distribution and Renal Handling

Approximately 85% of the body's phosphate is found in bone, where it plays a vital role in bone formation. The remainder is primarily in the ICF.

Phosphate is freely filtered by the glomeruli. 60% to 70% of filtered phosphate is reabsorbed in the PCT, and 15% is reabsorbed in the proximal straight tubule.

Na^+ -phosphate cotransporter located in the apical membrane facilitates the transportation of phosphate.

PTH facilitates phosphate excretion.



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Magnesium Distribution and Balance

Magnesium is abundant in the body and is involved in several physiological functions. These include protein synthesis, muscle and nerve function, etc.

60% of the body's Mg^{2+} is found in bone.

39% in the ICF, of which most is bound compounds like ATP.

1% in the ECF.

Mg^{2+} deficiency affects neuromuscular, cardiovascular, and gastrointestinal function.

Two thirds of the dietary Mg^{2+} is excreted in feces and one third in urine, with the kidneys playing a major role in the regulation of plasma Mg^{2+} .

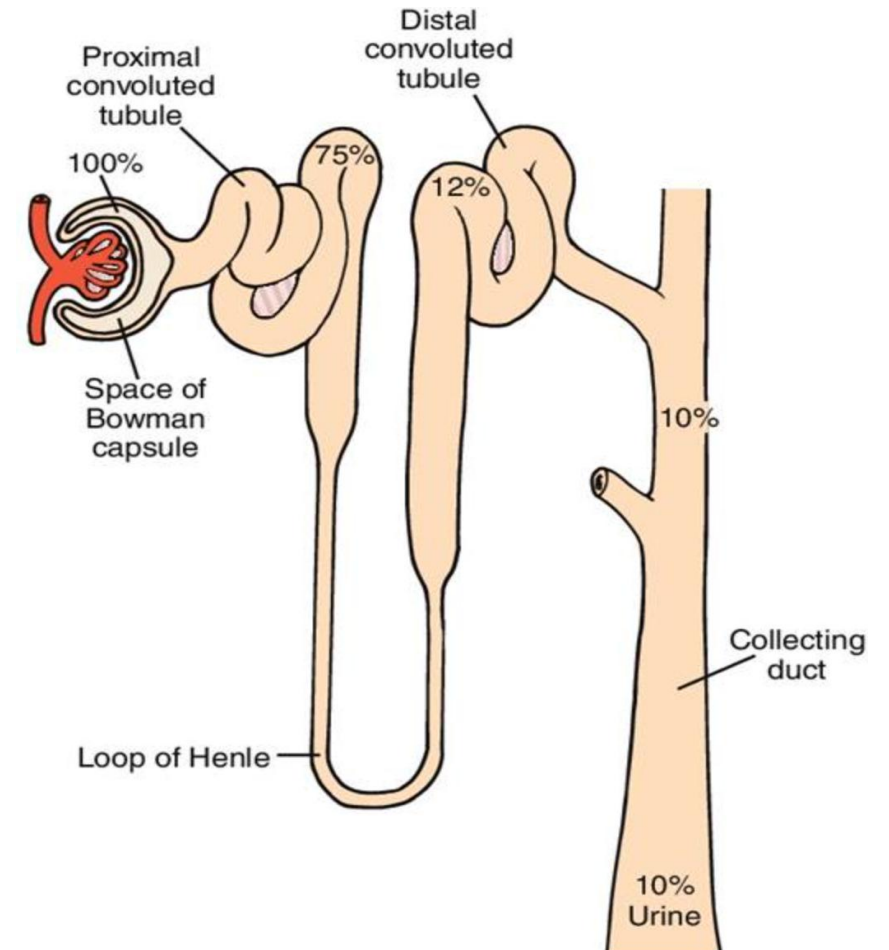
Renal Handling of Magnesium

About a quarter of the filtered Mg^{2+} is reabsorbed in the proximal convoluted tubule.

65% is reabsorbed in the loop of Henle. Reabsorption here, especially the thick ascending limb, is mainly passive via tight junctions and is driven by the lumen-positive potential.

Loop diuretics can reduce the reabsorption of Mg^{2+} .

The role of the distal tubule in the reabsorption Mg^{2+} is minor.



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References

- Medical Physiology: Principles for Clinical Medicine – Rodney A Rhoades & David A. Bell – 6th edition
- Medical Physiology– Boron & Burlap
- Guyton & Hall – 13th Edition

For question and clarifications please email:

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